Sales Data Analysis and Visualization Project

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Aspiring Data Analyst / Data Scientist

Tools Used:

Python

Pandas

Matplotlib

Seaborn

Jupyter Notebook

Introduction

This project focuses on analysing sales data to identify important patterns, trends, and insights. The project uses Python for data cleaning, data analysis, and data visualization. The goal of this project is to understand sales performance across different product categories, regions, and time periods and to generate meaningful business insights from the data.

Objective

The objectives of this project are:

To analyse overall sales performance

To identify top-selling products and categories

To analyse monthly and yearly sales trends

To identify top-performing regions and cities

To visualize sales data using charts and graphs

To generate business insights from the data

Dataset Description

The dataset used in this project contains sales transaction data. It includes order details, customer details, product categories, regions, and sales amount.

Important columns in the dataset include:

Order Date – Date when the order was placed

Ship Date – Date when the order was shipped

Category – Product category

Sub-Category – Product sub-category

Product Name – Name of the product

City – City of the customer

Region – Region of the customer

Sales – Sales amount

Data Cleaning

Data cleaning is an important step before analysis. In this project, the following data cleaning steps were performed:

Converted Order Date and Ship Date into datetime format

Extracted Month, Year, and Day from Order Date

Checked for missing values and handled them

Checked for duplicate records and removed them

Ensured correct data types for analysis

These steps ensured that the dataset was clean and ready for analysis.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the data and identify patterns and trends.

The following analysis was performed:

Total Sales Analysis

Sales by Year

Sales by Month

Sales by Category

Sales by Sub-Category

Top 10 Selling Products

Sales by Region

Top 10 Cities by Sales

Top Customers by Sales

Data Visualization

Data visualization was performed to represent the data in graphical format. The following charts were created:

Monthly Sales Bar Chart

Yearly Sales Bar Chart

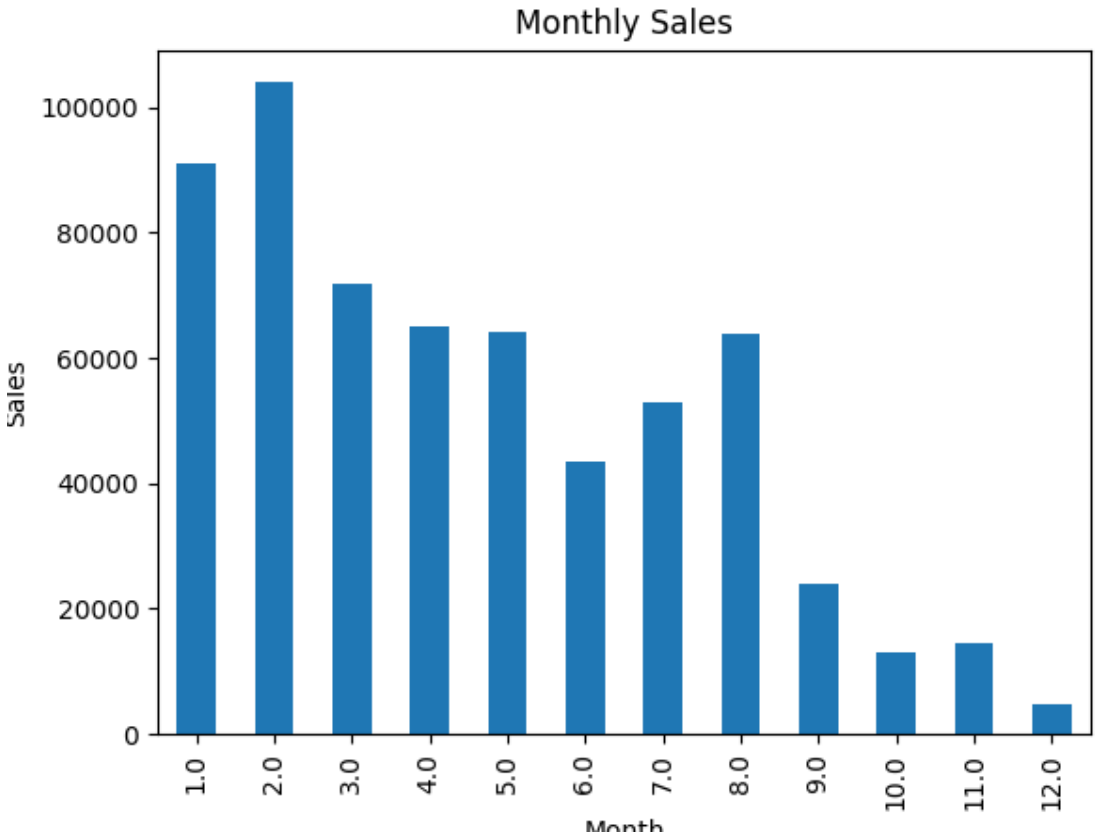
Top 10 Products Bar Chart

Sales by Category Pie Chart

Sales Trend Line Chart

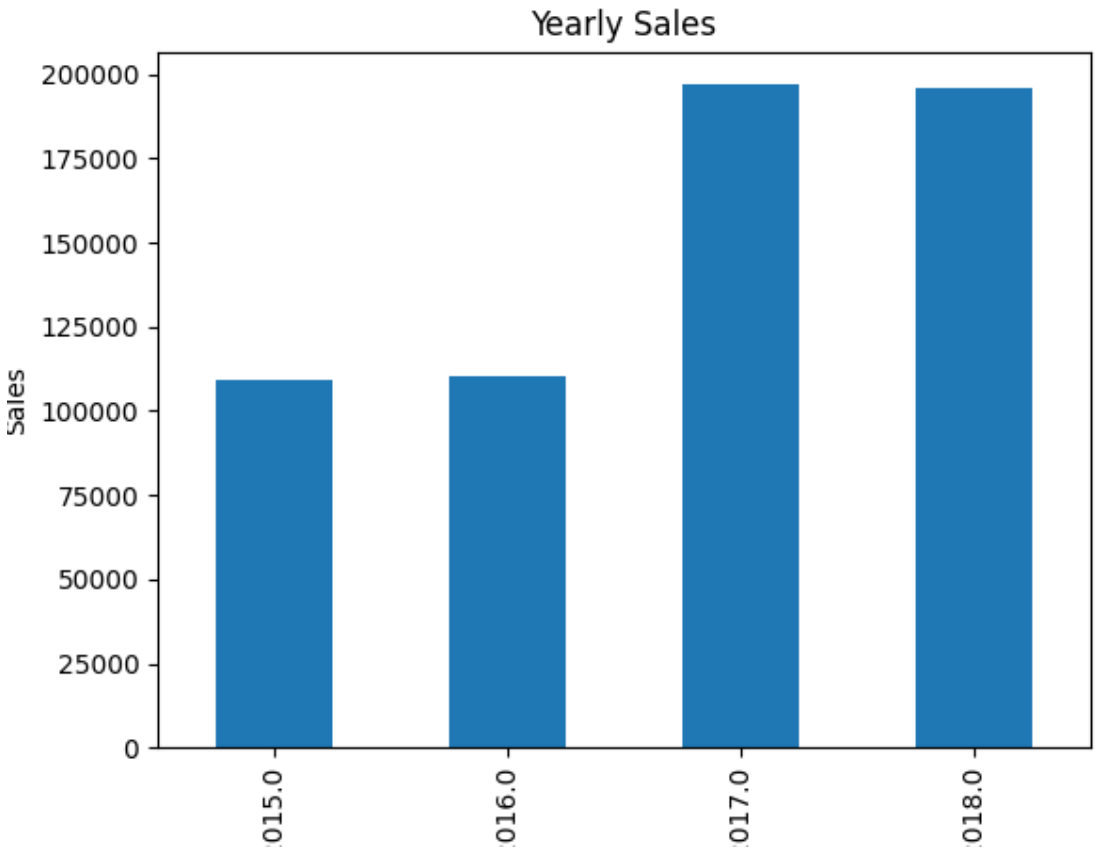
Correlation Heatmap

These visualizations help in understanding trends and patterns more easily.



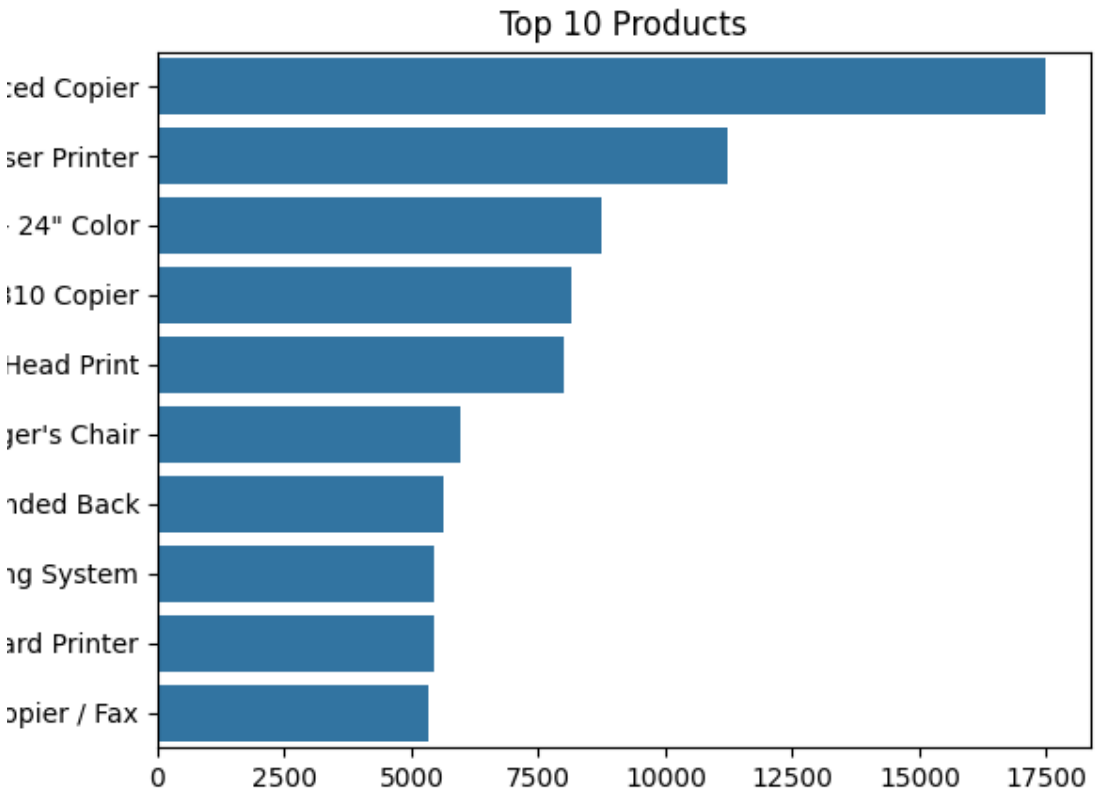
Monthly Sales Distribution

This chart shows monthly sales distribution. Some months have higher sales, indicating seasonal trends.



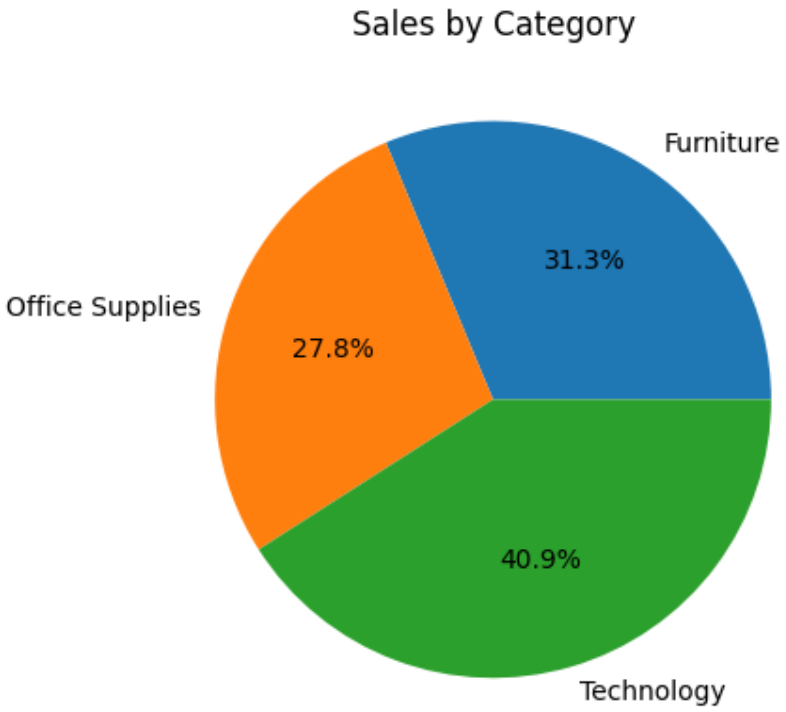
Yearly Sales

This chart shows total sales per year and overall growth trend.



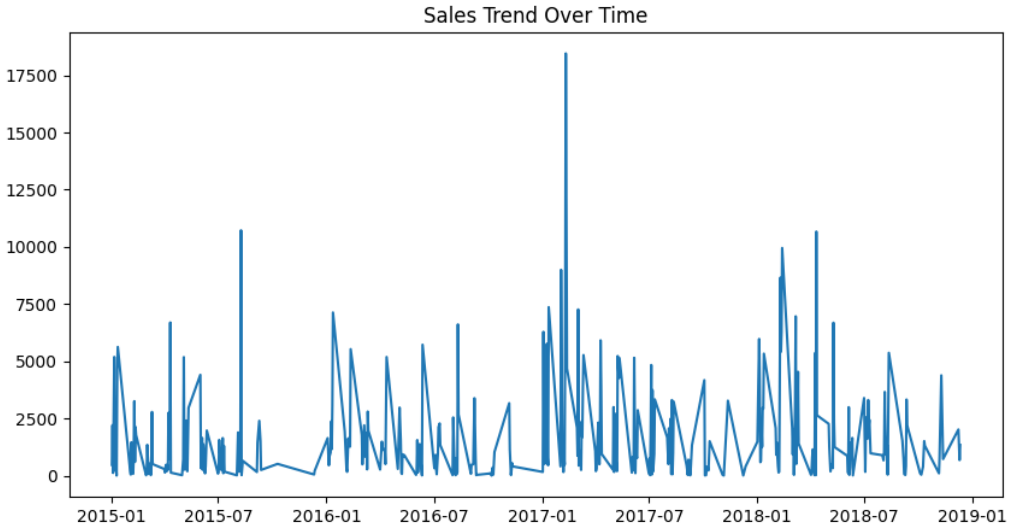
Top 10 Products by Sales

This chart shows the top-performing products based on sales.



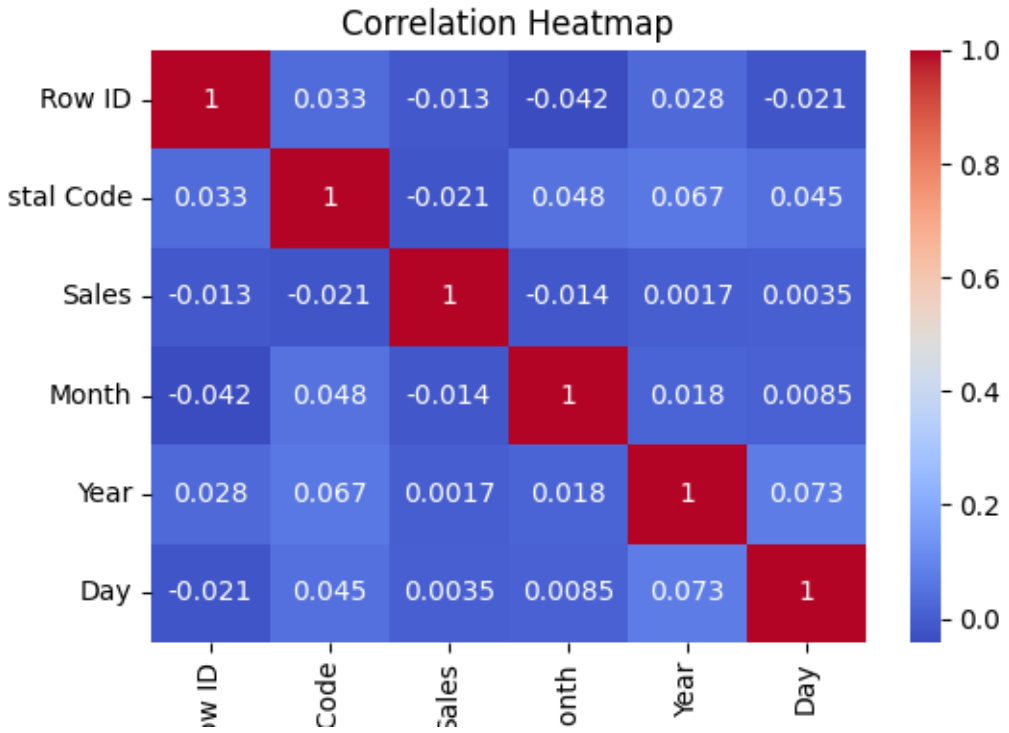
Sales by Category

This pie chart shows the contribution of each category to total sales.



Sales Trend Over Time

This line chart shows how sales change over time.



Correlation Heatmap

This heatmap shows correlation between numerical variables.

Business Insights

After analysing the sales data, the following business insights were identified:

Sales are higher in certain months, indicating seasonal demand.

The Technology category generates the highest sales.

A small number of products contribute to a large portion of total sales.

Some regions and cities perform better than others in terms of sales.

Top customers contribute significantly to total revenue.

These insights can help businesses improve marketing strategies, inventory planning, and sales strategies.

Conclusion

This project analysed sales data using Python. Data cleaning, data analysis, and data visualization techniques were used to understand sales performance. The project helped identify sales trends, top products, top regions, and customer behaviour. The insights gained from this analysis can help businesses make better decisions.

Future Work

This project can be further improved by:

Building a sales prediction model using machine learning

Performing customer segmentation

Creating an interactive dashboard using Plotly or Power BI

Analysing profit and discount data